Winstek Semiconductor Co., Ltd.（台星科）

Location: **Qionglin Township** and Hukou Township, Hsinchu County, Taiwan (the factory is in Qionglin)

Website: https://www.moneydj.com/kmdj/wiki/wikiviewer.aspx?keyid=b3c09f78-3120-42a7-a327-99e80bf1b240&utm_source

Video: https://www.youtube.com/watch?v=faxWjJ5lLUg

1. What did you learn from this resource?
We learned that 台星科 is a semiconductor company that mainly does chip packaging and testing. We also learned that the company is related to the development of AI and advanced semiconductor technology.

2. How can it help your PBL project?
This resource helps us understand what 台星科 does and its role in the semiconductor industry. We can use this information to introduce the company and learn more about how semiconductor companies affect Taiwan's technology industry.

Taiwan's Semiconductor Industry facing problems:

1. Crowding-Out Effects on Talent and Resources

The rapid expansion of the semiconductor and AI industries has created significant crowding-out effects on domestic electricity, land, basic resources, and high-quality technical talent. Traditional industries not only face pressure to transform and upgrade but also encounter extremely fierce salary competition and challenges in retaining talent in the recruitment market. At the same time, **Taiwanese IC design companies are under pressure from talent competition and product homogenization. In emerging, high-growth fields such as AI chip design and ASICs for edge computing, they must contend with direct competition from international giants.** Risks associated with the turnover of high-level talent and pressure from international salary bidding wars have become systemic challenges.

- Taiwan's Industrial Outlook for the Second Half of 2026: As AI Demand Expands, Supply Chains and Talent Strategy Become Key Competitive Factors
- The Boom and Structural Challenges Facing Taiwan's Semiconductor Industry in 2026]

Solution:

- Industry-Academia Collaboration: Partner with universities such as National Taiwan University, National Tsing Hua University, and National Chiao Tung University to identify and cultivate students early on, rather than waiting until after graduation to compete for talent.
- On-the-Job Training: Train existing employees to become higher-level talent, rather than relying entirely on poaching from outside the company.
- Recruiting Overseas Talent: Proactively bring talent back from abroad, rather than passively defending against poaching.

Significant Shortage of Semiconductor Talent: AI's Share of Hiring Continues to Rise, Assignments to the U.S. and Canada Reach Record Highs, and Opportunities for Non-Science Majors Increase

2. Production Capacity Cannot Keep Up with Demand

The issue facing Taiwan's semiconductor supply chain is no longer simply whether to expand production, but whether supply chain capacity can keep pace with demand; Taiwan's supply chain remains one of the bottlenecks in the global expansion of AI infrastructure. When TSMC estimated equipment demand for the first quarter of this year at the end of last year, the demand forecast—which was initially set at 1x—rose to 1.5x after just one quarter and was further raised to 1.9x in July of this year. The challenge facing the industry is for the entire ecosystem to support such massive growth within a mere six months.

- AI Demand Is Growing So Fast That Production Expansion Can't Keep Up—What's Next for Taiwan's Supply Chain?

Solution:

- Google, NVIDIA, and SK Hynix are evaluating a switch to Intel's EMIB packaging technology to avoid having all their orders backed up at TSMC's CoWoS production lines. Meanwhile, Singapore-based equipment manufacturer PBA is expanding its facilities in Singapore, Malaysia, and Taiwan to increase overall advanced packaging capacity.

3. Geopolitical and Raw Material Risks

The 2026 Middle East crisis has created invisible impacts on the semiconductor industry, ranging from energy security to microfabrication materials. Qatar, which accounts for more than one-third of global **helium production**, has seen its production facilities damaged in the conflict. Combined with disruptions to shipping through the Strait of Hormuz, this has triggered high alert levels across the global semiconductor industry; **helium is virtually irreplaceable in the cooling and lithography processes of chip manufacturing.** Furthermore, the supply crisis for hydrogen bromide is equally severe. **As a core material for plasma etching in advanced processes, a supply disruption lasting more than a quarter could force major memory manufacturers—and even some foundry lines—to choose between mature processes and high-margin AI chips when allocating production capacity.**

- Challenges and Restructuring of Taiwan's Semiconductor Industry Amid the 2026 Middle East Crisis and Geopolitical Trade Barriers

Solution:

- The same government statement confirmed that Taiwan's gas suppliers can already purchase helium from alternative international suppliers, specifically the US and Australia, and that the supply is stable.

4. Competitive Pressure from Taiwan and the U.S.

As Europe, the U.S., and Japan actively emphasize self-reliance in the semiconductor supply chain—with the **U.S. president even setting a goal of achieving a 50% global market share by 2029—there remains widespread concern that Taiwan's semiconductor industry may face the risk of production outflow** (although currently, as much as 83% of Taiwan's semiconductor production capacity remains domestically, and its core position has not yet been shaken).

- Taiwan's Semiconductor Core Remains Unshakable: ITRI Says Production Capacity Share Will Still Be Close to 80% in 2030
- Trump Impose 200% Tariffs on Chips, Vowing to Control 50% of the Global Semiconductor Industry by the End of His Term

Solution:

- **Taiwan's strategy**: trading some production capacity and mature processes for tariff concessions, while insisting on keeping advanced processes (3 nanometers and below) domestically—this is currently the core of its “Silicon Shield” strategy.
- Taiwanese officials continue to emphasize the need to diversify the economy, encourage growth in more industries, and strengthen national defense capabilities to counter geopolitical pressures, rather than simply putting all their eggs in the semiconductor basket.